

Final_Project

Blake Dee

2026-04-28

```
library(ggplot2)
library(readxl)
library(tidyverse)
```

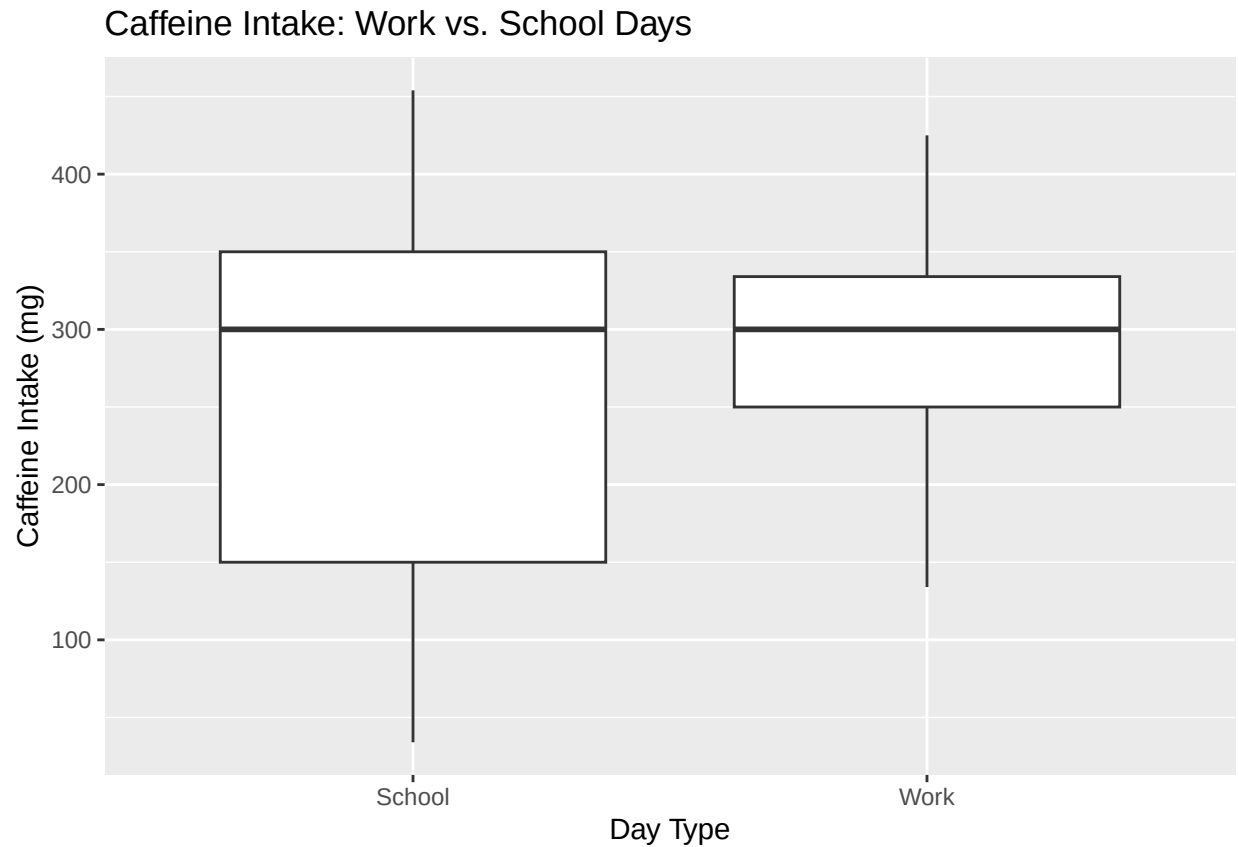
```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v lubridate  1.9.5      v tibble    3.3.1
## v purrr      1.2.1      v tidyr     1.3.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
caffeineData<-read_xlsx("C:/DATA211/DataSets/Caffeine Intake Analysis.xlsx")
```

ggplots

Boxplot

```
caffeineData%>%
  ggplot(aes(x=`Day Type`, y=`Caffeine Intake (mg)`))+geom_boxplot() +labs(title = "Caffeine Intake: Workday vs. Weekends")
```

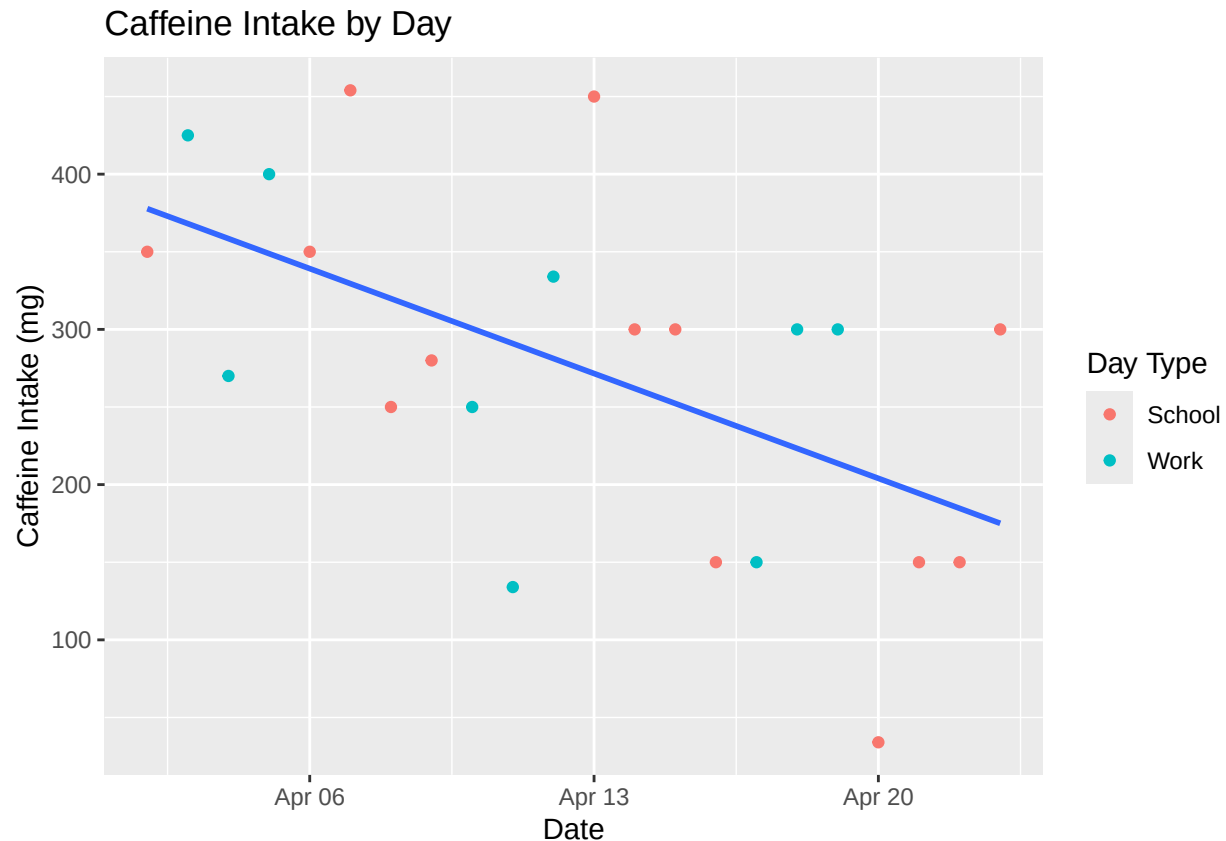


Here, we can see that the means are roughly the same. The only different is that there is a larger range of caffeine intake on school days than that of work days. Interestingly, despite the outlier of 34(mg), the outlier was not included as an outlier, but as a member of the range.

Scatter plot with regression

```
caffeineData%>%
  ggplot(aes(x=`DATE`,y=`Caffeine Intake (mg)`)) +geom_point(aes(color=`Day Type`)) +geom_smooth(method=

## 'geom_smooth()' using formula = 'y ~ x'
```



This better shows how the results of the distribution over time. I found it interesting to see the regression line sloping downwards throughout the observation days.

Hypothesis Test

μ_{school} : Mean amount of caffeine intake (mg) during school days μ_{work} : Mean amount of caffeine intake (mg) during work days

$H_0: \mu_{school} = \mu_{work}$ vs $H_a: \mu_{school} \neq \mu_{work}$ $\alpha = 0.05$

```
t.test(`Caffeine Intake (mg)` ~ `Day Type`, data = caffeineData, alternative="two.sided")
```

```
##
## Welch Two Sample t-test
##
## data: Caffeine Intake (mg) by Day Type
## t = -0.29879, df = 19.44, p-value = 0.7683
## alternative hypothesis: true difference in means between group School and group Work is not equal to
## 95 percent confidence interval:
## -113.21723 84.89245
## sample estimates:
## mean in group School mean in group Work
## 270.6154 284.7778
```

The p -value = 0.7683 which is greater than $\alpha = 0.05$. Based on a significance level of 5%, we fail to reject the H_0 . We do not have enough evidence to conclude that there is a difference between caffeine intake on work days versus school days.